

'Assignment'

Amirum Nahin

ID: 23201416

Sec: 09

Course: CSE427

Semester: Spring 2026

Date: 15/05/2026

Ans. to the question no. 2

from the given neural network,

assuming the loss function is sum of squared error,

$$L = \sum_{i=1}^n (ob - y)^2$$

Now, given,

weights, $w = 0.5$

biases, $b = 0.5$

$$x_1 = 1$$

$$x_2 = 1$$

$$ob_6 = 1$$

$$ob_7 = 0$$

from forward propagation,

$$Y_3 = \sigma(x_1 w_{13} + x_2 w_{23} + b_3) = 0.818$$

$$Y_4 = \sigma(x_1 w_{14} + x_2 w_{24} + b_4) = 0.818$$

$$Y_5 = \sigma(x_1 w_{15} + x_2 w_{25} + b_5) = 0.818$$

$$Y_6 = \sigma(Y_3 w_{36} + Y_4 w_{46} + Y_5 w_{56} + b_6) = 0.849$$

$$Y_7 = \sigma(Y_3 w_{37} + Y_4 w_{47} + Y_5 w_{57} + b_7) = 0.849$$

Now,

$$\frac{\partial L}{\partial Y_6} = \frac{\partial}{\partial Y_6} \{ (ob_6 - Y_6)^2 \}$$

$$= 2 (ob_6 - Y_6) (-1)$$

$$\begin{aligned}
 &= -2(0.66 - Y_6) \\
 &= -2(1 - 0.849) \\
 &= -0.302
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 \frac{\partial L}{\partial Y_7} &= -2(0.67 - Y_7) \\
 &= -2(0 - 0.849) \\
 &= 1.698
 \end{aligned}$$

and,

$$\begin{aligned}
 \frac{\partial L}{\partial Y_3} &= \frac{\partial L}{\partial Y_6} \cdot \frac{\partial Y_6}{\partial Y_3} + \frac{\partial L}{\partial Y_7} \cdot \frac{\partial Y_7}{\partial Y_3} \\
 &= -0.302 \times Y_6' \cdot w_{36} + 1.698 \cdot Y_7' \cdot w_{37} \\
 &= -0.302 \times 0.849(1 - 0.849) \cdot 0.5 + 1.698 \times 0.849(1 - 0.849) \times 0.5 \\
 &= 0.0895
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial L}{\partial Y_4} &= \frac{\partial L}{\partial Y_6} \cdot \frac{\partial Y_6}{\partial Y_4} + \frac{\partial L}{\partial Y_7} \cdot \frac{\partial Y_7}{\partial Y_4} \\
 &= -0.302 \times Y_6(1 - Y_6) \cdot w_{46} + 1.698 \times Y_7(1 - Y_7) \cdot w_{47} \\
 &= 0.0895
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial L}{\partial Y_5} &= \frac{\partial L}{\partial Y_6} \cdot \frac{\partial Y_6}{\partial Y_5} + \frac{\partial L}{\partial Y_7} \cdot \frac{\partial Y_7}{\partial Y_5} \\
 &= -0.302 \times Y_6(1 - Y_6) \cdot w_{56} + 1.698 \times Y_7(1 - Y_7) \cdot w_{57} \\
 &= 0.0895
 \end{aligned}$$

Now,

$$\begin{aligned}w_{13} &= w_{13} - \eta \cdot \frac{\partial L}{\partial w_{13}} \\(\text{new}) & \quad (\text{old}) \\&= 0.5 - 0.1 \times \left(\frac{\partial L}{\partial y_3} \cdot \frac{\partial y_3}{\partial w_{13}} \right) \\&= 0.5 - 0.1 \times \left(0.0895 \cdot y_3(1-y_3) \cdot x_1 \right) \\&= 0.5 - 0.1 \times \left(0.0895 \times 0.818(1-0.818) \cdot 1 \right) \\&= 0.4986\end{aligned}$$

and,

$$\begin{aligned}w_{24} &= w_{24} - \eta \frac{\partial L}{\partial w_{24}} \\(\text{new}) & \quad (\text{old}) \\&= 0.5 - 0.1 \left(\frac{\partial L}{\partial y_4} \cdot \frac{\partial y_4}{\partial w_{24}} \right) \\&= 0.5 - 0.1 \left(0.0895 \cdot y_4(1-y_4) \cdot x_2 \right) \\&= 0.4986\end{aligned}$$

and,

$$\begin{aligned}w_{23} &= w_{23} - \eta \frac{\partial L}{\partial w_{23}} \\(\text{new}) & \quad (\text{old}) \\&= 0.5 - 0.1 \left(\frac{\partial L}{\partial y_3} \cdot \frac{\partial y_3}{\partial w_{23}} \right) \\&= 0.5 - 0.1 \left(0.0895 \cdot y_3(1-y_3) \cdot x_2 \right) \\&= 0.4986\end{aligned}$$

$$\begin{aligned}
 w_{14} &= w_{14} - \eta \frac{\partial L}{\partial w_{14}} \\
 (\text{new}) & (\text{old}) \\
 &= 0.5 - 0.1 \left(\frac{\partial L}{\partial y_4} \cdot \frac{\partial y_4}{\partial w_{14}} \right) \\
 &= 0.5 - 0.1 \left(0.0895 \cdot y_4 (1 - y_4) \cdot 1 \right) \\
 &= 0.4986
 \end{aligned}$$

$$\begin{aligned}
 b_3 &= b_3 - \eta \frac{\partial L}{\partial b_3} \\
 (\text{new}) & (\text{old}) \\
 &= 0.5 - 0.1 \times \left(\frac{\partial L}{\partial y_3} \cdot \frac{\partial y_3}{\partial b_3} \right) \\
 &= 0.5 - 0.1 \times \left(0.0895 \times y_3 (1 - y_3) \right) \\
 &= 0.4986
 \end{aligned}$$

$$\begin{aligned}
 b_4 &= b_4 - \eta \frac{\partial L}{\partial b_4} \\
 (\text{new}) & (\text{old}) \\
 &= 0.5 - 0.1 \left(\frac{\partial L}{\partial y_4} \cdot \frac{\partial y_4}{\partial b_4} \right) \\
 &= 0.5 - 0.1 \left(0.0895 \times 0.818 (1 - 0.818) \cdot 1 \right) \\
 &= 0.4986
 \end{aligned}$$

$$\begin{aligned}
 b_5 &= b_5 - \eta \frac{\partial L}{\partial b_5} \\
 (\text{new}) & (\text{old}) \\
 &= 0.5 - 0.1 \left(\frac{\partial L}{\partial y_5} \cdot \frac{\partial y_5}{\partial b_5} \right) \\
 &= 0.5 - 0.1 \left(0.0895 \cdot y_5 (1 - y_5) \cdot 1 \right) \\
 &= 0.4986
 \end{aligned}$$

finally, the updated values are,

$$w_{13} = 0.4986$$

$$w_{23} = 0.4986$$

$$w_{14} = 0.4986$$

$$w_{24} = 0.4986$$

$$b_3 = 0.4986$$

$$b_4 = 0.4986$$

$$b_5 = 0.4986$$

(Ans.)

Ans. to the que. no-03

We know,

$$TPR = \text{Sensitivity} = \frac{TP}{TP + FN}$$

$$FPR = 1 - \text{specificity} = \frac{FP}{FP + TN}$$

Now, for threshold = 1,

$$TPR = 0$$

$$FPR = 0$$

for threshold = 0.965,

$$TPR = \frac{2}{2+4} = 0.33$$

$$FPR = 0$$

for threshold = 0.975

$$TPR = \frac{1}{1+5} = 0.167$$

$$FPR = 0$$

for threshold = 0.95,

$$TPR = \frac{3}{3+3} = 0.5$$

$$FPR = 0$$

for threshold = 0.87,

$$TPR = \frac{3}{3+3} = 0.5$$

$$FPR = \frac{1}{1+3} = 0.25$$

for threshold = 0.50,

$$TPR = \frac{4}{4+2} = 0.67$$

$$FPR = \frac{1}{1+3} = 0.25$$

for threshold = 0.75,

$$TPR = \frac{4}{4+2} = 0.67$$

$$FPR = \frac{1}{1+3} = 0.25$$

for threshold = 0.20,

$$TPR = \frac{6}{6+0} = 1$$

$$FPR = \frac{1}{1+3} = 0.25$$

for threshold = 0.15,

$$TPR = \frac{6}{6+0} = 1$$

$$FPR = \frac{2}{2+2} = 0.5$$

for threshold = 0.10,

$$TPR = \frac{6}{6+0} = 1$$

$$FPR = \frac{3}{3+1} = 0.75$$

for threshold = 0.05,

$$TPR = \frac{6}{6+0} = 1$$

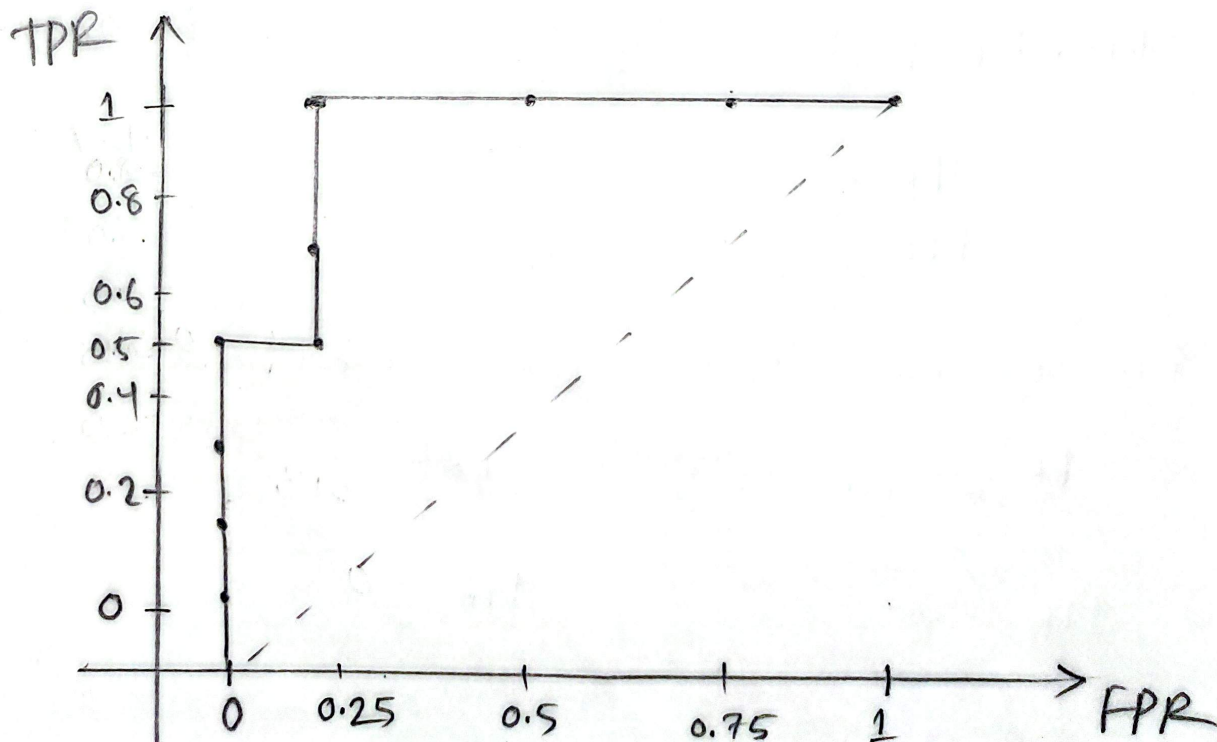
$$FPR = \frac{4}{4+0} = 1$$

for threshold = 0,

$$TPR = \frac{6}{6+0} = 1$$

$$FPR = \frac{4}{4+0} = 1$$

The ROC curve,



To get 100% True positive rate we have to set the probability threshold at 0.20 or lower.

And to get 0% false positive we have to set the probability threshold at 0.95 or higher.

Ans. to the que. no-4

from the given table,

for class A,

$$\text{False positive (FP)} = 50 + 200 = 250$$

$$\text{False Negative (FN)} = 100 + 100 = 200$$

$$\begin{aligned}\text{sensitivity} &= \frac{TP}{TP + FN} \\ &= \frac{800}{800 + 200} \\ &= 0.8\end{aligned}$$

$$\begin{aligned}\text{Specificity} &= \frac{TN}{FP + TN} \\ &= \frac{900 + 50 + 100 + 700}{250 + 900 + 50 + 100 + 700} \\ &= 0.875\end{aligned}$$

$$\text{precision} = \frac{TP}{TP + FP}$$

$$= \frac{800}{800+250}$$

$$= 0.762$$

$$\text{Recall} = \text{Sensitivity} = 0.8$$

for class B,

$$\text{False Positive (FP)} = 100 + 100 = 200$$

$$\text{False Negative (FN)} = 50 + 50 = 100$$

$$\begin{aligned} \text{sensitivity} &= \frac{TP}{TP + FN} \\ &= \frac{900}{900 + 100} \\ &= 0.9 \end{aligned}$$

$$\begin{aligned} \text{specificity} &= \frac{TN}{FP + TN} \\ &= \frac{1800}{1800 + 200} \\ &= 0.9 \end{aligned}$$

$$\begin{aligned} \text{precision} &= \frac{TP}{TP + FP} \\ &= \frac{900}{900 + 200} \\ &= 0.818 \end{aligned}$$

$$\text{Recall} = \text{Sensitivity} = 0.9$$

for class C,

$$\text{False Positive (FP)} = 100 + 50 = 150$$

$$\text{False Negative (FN)} = 200 + 100 = 300$$

$$\begin{aligned}\text{Sensitivity} &= \frac{TP}{TP + FN} \\ &= \frac{700}{700 + 300} \\ &= 0.7\end{aligned}$$

$$\begin{aligned}\text{specificity} &= \frac{TN}{FP + TN} \\ &= \frac{1850}{1850 + 150} \\ &= 0.925\end{aligned}$$

$$\begin{aligned}\text{precision} &= \frac{TP}{TP + FP} \\ &= \frac{700}{700 + 150} \\ &= 0.824\end{aligned}$$

$$\text{Recall} = \text{Sensitivity} = 0.7$$

(Ans.)

Ans to the que no- 08

Given embedding values,

"We"

$$\begin{bmatrix} 1.87 & 0.09 \end{bmatrix}$$

"Go"

$$\begin{bmatrix} -0.78 & 0.27 \end{bmatrix}$$

Now, for "We", positional embedding values are,

$$\text{pos}(0, 0) = \sin \frac{0}{10000} \left(\frac{0}{2} \right) = 0$$

$$\text{pos}(0, 1) = \cos \frac{0}{10000} \left(\frac{0}{2} \right) = 1$$

for "Go"

$$\text{pos}(1, 0) = \sin \frac{1}{10000} \left(\frac{0}{2} \right) = 0.841$$

$$\text{pos}(1, 1) = \cos \frac{1}{10000} \left(\frac{0}{2} \right) = 0.54$$

embedding values of We and Go after adding positional embeddings are,

We

$$\begin{bmatrix} 1.87 & 1.09 \end{bmatrix}$$

Go

$$\begin{bmatrix} 0.061 & 0.81 \end{bmatrix}$$

Now, QKV matrices for W_e and G_o ,

$$Q_W = \begin{bmatrix} 1.87 & 1.09 \end{bmatrix} \begin{bmatrix} 1.1 & 0.6 \\ -2.8 & 2.4 \end{bmatrix} = \begin{bmatrix} -0.995 & 3.738 \end{bmatrix}$$

$$K_W = \begin{bmatrix} 1.87 & 1.09 \end{bmatrix} \begin{bmatrix} -1.7 & 0.5 \\ -1.5 & 0.9 \end{bmatrix} = \begin{bmatrix} -4.814 & 1.916 \end{bmatrix}$$

$$V_W = \begin{bmatrix} 1.87 & 1.09 \end{bmatrix} \begin{bmatrix} 1.5 & -1.0 \\ -0.3 & -0.2 \end{bmatrix} = \begin{bmatrix} 2.478 & -2.088 \end{bmatrix}$$

$$Q_G = \begin{bmatrix} 0.061 & 0.81 \end{bmatrix} \begin{bmatrix} 1.1 & 0.6 \\ -2.8 & 2.4 \end{bmatrix} = \begin{bmatrix} -2.201 & 1.981 \end{bmatrix}$$

$$K_G = \begin{bmatrix} 0.061 & 0.81 \end{bmatrix} \begin{bmatrix} -1.7 & 0.5 \\ -1.5 & 0.9 \end{bmatrix} = \begin{bmatrix} -1.319 & 0.759 \end{bmatrix}$$

$$V_G = \begin{bmatrix} 0.061 & 0.81 \end{bmatrix} \begin{bmatrix} 1.5 & -1.0 \\ -0.3 & -0.2 \end{bmatrix} = \begin{bmatrix} -0.151 & -0.223 \end{bmatrix}$$

Now, to calculate self attention of W_e ,

$$\begin{aligned} Q_W \cdot K_W &= \begin{bmatrix} -0.995 & 3.738 \end{bmatrix} \cdot \begin{bmatrix} -4.814 & 1.916 \end{bmatrix} \\ &= 11.952 \end{aligned}$$

$$\begin{aligned} Q_W \cdot K_G &= \begin{bmatrix} -0.995 & 3.738 \end{bmatrix} \cdot \begin{bmatrix} -1.319 & 0.759 \end{bmatrix} \\ &= 4.149 \end{aligned}$$

again, self-attention of G_0 ,

$$Q_0 \times K_0 = \begin{bmatrix} -2.201 & 1.981 \end{bmatrix} \cdot \begin{bmatrix} -1.319 & 0.759 \end{bmatrix} \\ = 4.407$$

$$Q_0 \times K_W = \begin{bmatrix} -2.201 & 1.981 \end{bmatrix} \cdot \begin{bmatrix} -4.814 & 1.916 \end{bmatrix} \\ = 14.391$$

Now, applying softmax to these values,

for W ,

$$\text{softmax}(11.952, 4.149) \\ = \begin{bmatrix} 0.999 & 0.000408 \end{bmatrix}$$

for G_0 ,

$$\text{softmax}(4.407, 14.391) \\ = \begin{bmatrix} 0.0000461 & 0.999 \end{bmatrix}$$

scaling the value matrix with these values,

for W ,

$$\begin{bmatrix} 2.478 & -2.088 \end{bmatrix} \times 0.999 = \begin{bmatrix} 2.475 & -2.085 \end{bmatrix} \\ \begin{bmatrix} -0.151 & -0.223 \end{bmatrix} \times 0.00048 = \begin{bmatrix} -7.248 \times 10^{-5} & -1.07 \times 10^{-4} \end{bmatrix}$$

adding these two values,

$$\begin{bmatrix} 2.476 & -2.086 \end{bmatrix}$$

Similarly for G_0 ,

$$\begin{bmatrix} -0.151 & -0.223 \end{bmatrix} \times 0.0000461 = \begin{bmatrix} -6.96 \times 10^{-6} & -1.02 \times 10^{-5} \end{bmatrix}$$

$$\begin{bmatrix} 2.478 & -2.088 \end{bmatrix} \times 0.999 = \begin{bmatrix} 2.475 & -2.087 \end{bmatrix}$$

adding these values,

$$\begin{bmatrix} 2.475 & -2.087 \end{bmatrix}$$

Calculating residual connection for W_e ,

$$\begin{bmatrix} 2.476 & -2.086 \end{bmatrix} + \begin{bmatrix} 1.87 & 1.09 \end{bmatrix} = \begin{bmatrix} 4.346 & -0.996 \end{bmatrix}$$

calculating residual connection for G_0 ,

$$\begin{bmatrix} 2.475 & -2.087 \end{bmatrix} + \begin{bmatrix} 0.061 & 0.81 \end{bmatrix} = \begin{bmatrix} 2.536 & -1.277 \end{bmatrix}$$

This is our encoder's output for W_e and G_0 .

Now, given embedded values for decoder's side $\langle \text{Eos} \rangle$,

$$\begin{bmatrix} 2.7 & -1.34 \end{bmatrix}$$

Now, Let's find the QVK matrices for cross attention,

$$QE = \begin{bmatrix} 2.7 & -1.34 \end{bmatrix} \begin{bmatrix} 1.5 & -0.3 \\ 0.3 & -1.0 \end{bmatrix} = \begin{bmatrix} 3.648 & 0.53 \end{bmatrix}$$

$$KW = \begin{bmatrix} 4.346 & -0.996 \end{bmatrix} \begin{bmatrix} -1.1 & 0.3 \\ -1.5 & -0.8 \end{bmatrix} = \begin{bmatrix} -3.286 & 2.101 \end{bmatrix}$$

$$VW = \begin{bmatrix} 4.346 & -0.996 \end{bmatrix} \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & -0.5 \end{bmatrix} = \begin{bmatrix} 5.976 & 3.106 \end{bmatrix}$$

$$KG = \begin{bmatrix} 2.536 & -1.277 \end{bmatrix} \begin{bmatrix} -1.1 & 0.3 \\ -1.5 & -0.8 \end{bmatrix} = \begin{bmatrix} -0.874 & 1.782 \end{bmatrix}$$

$$VG = \begin{bmatrix} 2.536 & -1.277 \end{bmatrix} \begin{bmatrix} 1.1 & 0.6 \\ -1.2 & -0.5 \end{bmatrix} = \begin{bmatrix} 4.322 & 2.160 \end{bmatrix}$$

So,

$$QE \cdot KW = \begin{bmatrix} 3.648 & 0.53 \end{bmatrix} \cdot \begin{bmatrix} -3.286 & 2.101 \end{bmatrix} \\ = -10.874$$

$$QE \cdot KG = \begin{bmatrix} 3.648 & 0.53 \end{bmatrix} \begin{bmatrix} -0.874 & 1.782 \end{bmatrix} \\ = -2.244$$

applying softmax,

$$\text{softmax}(-10.874, -2.244)$$

$$= [0.000178 \quad 0.999]$$

scaling these values with value matrix,

$$[5.976 \quad 3.106] \times 0.000178 = [1.06 \times 10^{-3} \quad 5.53 \times 10^{-4}]$$

$$[4.322 \quad 2.160] \times 0.999 = [4.318 \quad 2.158]$$

adding these value up,

$$= [4.318 \quad 2.158]$$

Then adding the positional embedding values,

$$[4.318 \quad 2.158] + [2.7 \quad -1.34] = [7.018 \quad 0.818]$$

So, our final answer,

$$[7.018 \quad 0.818]$$

(Ans.)